

Prem Patel

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EDUCATION

University of Waterloo

Waterloo, ON

BCS (Computer Science) | GPA: 3.7

Expected Apr. 2027

- Data Structures & Algorithms, Data-Intensive Distributed Computing, Databases, Operating Systems, Compilers

EXPERIENCE

Tesla

May 2026 – Aug. 2026

Software Engineer Intern, Distributed Systems

Palo Alto, CA

- Developed distributed **Scala services** using **Akka actors** to aggregate and process real-time data from **millions of Powerwall IoT devices**, deploying through **Argo CD** and **Kubernetes**
- Built a **full-stack analytics application** to display delivery performance of Powerwall 3 residential batteries enrolled in **Virtual Power Plant** programs; reduced response time for VPP partner escalations by **90%**
- Built an **ETL-based prototype** in under a week, then incorporated user feedback through iterative releases, and evolved it into a **production-ready architecture** with a **gRPC API** and **indexed PostgreSQL schema**

Kinaxis

Jan. 2025 – Apr. 2026

Software Engineer Intern, Embedded Algorithms

Ottawa, ON

- Built end-to-end features in a **distributed C++ server** with **TypeScript (Node.js)** + **Java** agents/workers over persistent **WebSockets**, improving reliability, latency, and throughput under load
- Shipped a **VS Code extension** to author, publish, and run/debug user-defined **embedded algorithms** (EA) with breakpoints, reducing developer iteration time by **40%**
- Launched **BYO-Model** with a solver-agnostic, **async job pipeline** to run optimization models at scale
- Designed thread-safe CPU/memory **guardrails** in **C++** using **RAII semaphores** to prevent noisy-neighbor resource spikes in **multi-tenant** clusters; cut lock contention by **32%**
- Implemented an **IPC state handler** with retry scheduling and **disconnection detection**, improving **fault tolerance** and preventing cascading failure logs after worker/agent disconnects
- Standardized **structured logging** + metrics integrated with Datadog dashboards to attribute CPU usage per customer, enabling transparent **consumption-based pricing** generating **\$1M+** annual revenue
- Improved response serialization with **FlatBuffers** to support **25%** larger payloads and eliminated silent **Node.js** crashes, increasing throughput and reliability under load
- Deployed EA services on **Kubernetes** via **Docker** + **Helm** (**Azure/GCP**) and presented **17** cross-team demos

AutoTrader

Jan. 2024 – May 2024

Software Engineer Intern, Data Automation

Toronto, ON

- Developed **Python** services processing OEM feeds for **30+ automotive brands**; reduced end-to-end processing time by **8x** by restructuring transformations and batching (Pandas + REST API integrations)
- Built **Azure Pipelines** automation to generate and deliver monthly reports for **150+ dealerships**, integrating **Redash** queries and enforcing consistent schemas/validation
- Automated PDF/table ingestion and validation with **Tabula**, reducing manual cleanup by **5 hrs** per refresh cycle and improving reporting accuracy by up to **85%**

PROJECTS

🔗 Ledger-First Transaction System | TypeScript, Node.js, PostgreSQL, Prisma

Jan. 2026

- Designed an immutable double-entry ledger (derived balances) with async trade execution; ensured auditability and replay-safe retries via transactional idempotency keys and **SYSTEM** counterparties enforcing financial invariants

TECHNICAL SKILLS

Languages: C++, Java, Scala, Python, TypeScript (JavaScript), Go, SQL, Bash, C, C#

Frameworks/Libs: Spring Boot (DI, Filters, MDC, WebSockets), FastAPI, Fastify, Prisma, React, Boost, FlatBuffers

Dev Tools: Git, VS/VS Code, Node.js, Docker, Kubernetes, PostgreSQL, Redis, Jenkins, Datadog, GCP, Azure, AWS